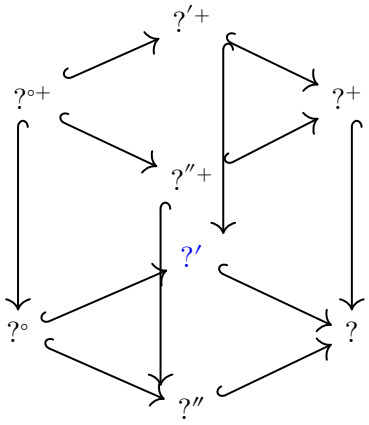
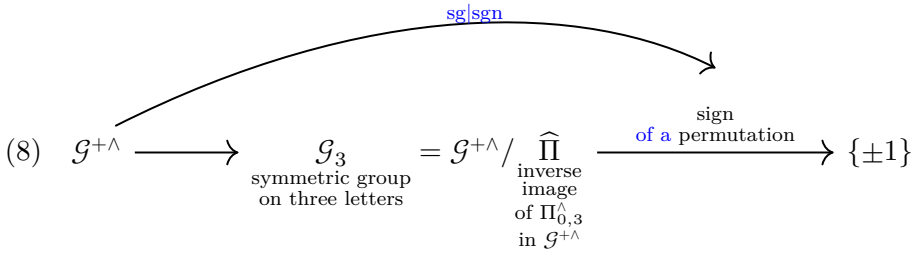
[scan 562]



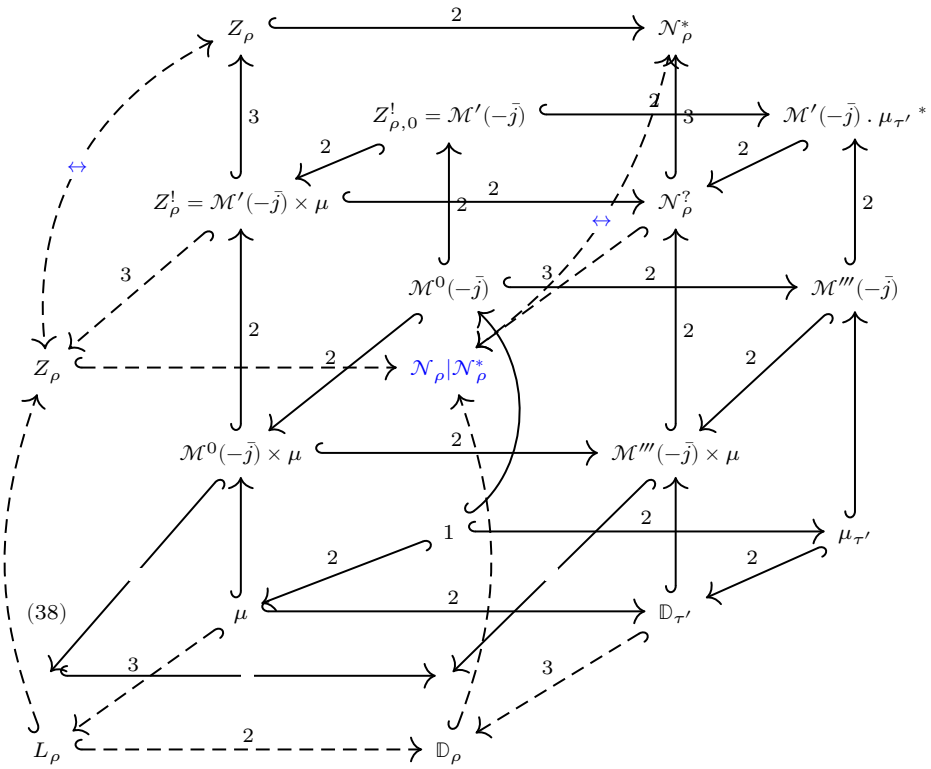
[scan 573]

$$(3)\quad \mathcal{M}^{\sim}_{0,3} \longrightarrow \mathrm{Aut}(\mathcal{M}) \xrightarrow[\mathcal{G}^{+\wedge}=\mathrm{Sl}(2,\mathbb{Z})^{\wedge}]{\text{restriction to}} \mathrm{Aut}(\widehat{\mathcal{G}}^{+})$$

[scan 574]

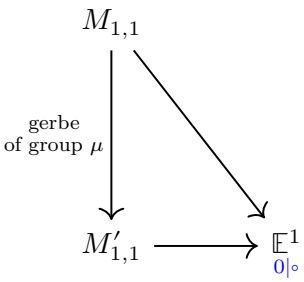

sg|sgn
(8)
$\mathcal{G}^{+\wedge}$
$\mathcal{G}_3$
symmetric group
on three letters
$= \mathcal{G}^{+\wedge} / \widehat{\Pi}$
inverse
image
of $\Pi^{\wedge}_{0,3}$
in $\mathcal{G}^{+\wedge}$
sign
of a permutation
$\{\pm 1\}$

[scan 638]


Z_ρ
$\mathcal{N}^*_\rho$
$Z^!_{\rho,0} = \mathcal{M}'(-\bar{j})$
$\mathcal{M}'(-\bar{j}) \cdot \mu_{\tau'}{}^*$
$Z^!_\rho = \mathcal{M}'(-\bar{j}) \times \mu$
$\mathcal{N}^?_\rho$
$\mathcal{M}^0(-\bar{j})$
$\mathcal{M}'''(-\bar{j})$
Z_ρ
$\mathcal{N}_\rho | \mathcal{N}^*_\rho$
$\mathcal{M}^0(-\bar{j}) \times \mu$
$\mathcal{M}'''(-\bar{j}) \times \mu$
$\mu_{\tau'}$
(38)
μ
$\mathbb{D}_{\tau'}$
L_ρ
$\mathbb{D}_\rho$

[scan 628]

moduli scheme of elliptic curves



moduli scheme of elliptic curves mod symmetry

coarse moduli scheme